

Biotechnology Development Facility Seed Grant Proposal

Sequencing Alfalfa Germplasm Sources to Evaluate a New Genotyping Method for Genomic Prediction and Selection

Principal Investigator: Kelly Robbins, Plant Breeding, SIPS, krr73@cornell.edu

Key Personnel: Nicholas Santantonio, Plant Breeding, SIPS, ns722@cornell.edu

The proposed sequencing data will be used as a proof of concept genotyping experiment for a grant to be submitted to USDA-AFRI in March 2020. In line with Cornell's initiative for advancement of digital agriculture, that research will focus on using remote sensing to understand genotype specific growth curves that lead to genotypic differences in yield. Using genetic relationships estimated from genotype data, we will build genomic prediction models that can model crop growth through time to allow for remote phenotyping. Alfalfa, a forage crop, was chosen because the harvested product is the total aboveground biomass, and therefore can be imaged directly from an aerial unmanned vehicle.

The challenge of using alfalfa is that a variety (or cultivar) is a population, instead of a genetically distinct individual. Therefore, genotyping a single individual from the population is insufficient to capture the allelic diversity of a variety or breeding population. The research proposed here will evaluate a new genotyping strategy for outcrossing crop species, where DNA from many (≥ 100) individuals will be used to estimate allele frequencies in that population. Allele counts within each variety/breeding population, as opposed to allele counts within each individual, will be used to estimate genetic relationships among populations.

The plant materials chosen for whole genome sequencing (WGS) are from the nine historic germplasm sources of alfalfa (Barnes et al. 1977; Segovia-Lerma et al. 2003), representing much of the genetic diversity of the crop. The proposed sequencing will allow us to bulk genotype these populations and evaluate genomic predictive ability of such a strategy using yield data from a diallel cross of the nine populations ($n=36$) previously phenotyped in Las Cruces, NM (Segovia-Lerma et al. 2004).

We will be providing these materials to Breeding Insight (BI), who are currently working to build a high-throughput genotyping platform for alfalfa. This panel of alfalfa accessions is complementary to a diversity panel of individuals from improved North American breeding germplasm currently undergoing skim sequencing within BI. Haplotype information between these two panels will allow for targeting informative regions of the alfalfa genome to aid in the construction of high-throughput sequence-based marker platform that can capture the allelic diversity across improved germplasm and sources of new genetic variability.

We were recently awarded a U.S. Alfalfa Farmer Research Initiative grant to genotype and image Cornell alfalfa breeding populations in the field. Demonstration of this genotyping method in a highly diverse background as well as more narrow adapted germplasm will allow us to evaluate the efficacy of the proposed genotyping method for pursuing digital agriculture solutions in the USDA-AFRI grant proposal.

Scientific Objectives and Rationale

Alfalfa (*Medicago sativa* subsp. *sativa*), known as the “Queen of forages” for its high protein content, is the most important forage crop in US, outproduced only by maize, soy and wheat. Enrichment of beneficial alleles in alfalfa is exceedingly difficult due to high inbreeding depression and the autotetraploid nature of the crop. As an obligate out-crosser, alfalfa must always be bred on a population level, where varieties are developed and released as “synthetic” populations that avoid inbreeding and take advantage of population level heterosis. This has limited implementation of marker-based selection because large numbers of individuals must be genotyped and inter-mated to avoid inbreeding in future generations. Single individuals are not representative of a variety as a whole, and genotyping many individuals from each variety is costly and restrictive.

While these limitations have restrained implementation of marker-based selection in alfalfa, they also represent a unique opportunity. Because much of quantitative genetics and selection theory hinges on population level parameters, the current machinery can easily be adapted to breeding on a population level. Tissue from many individuals within a variety or breeding population will be bulked for DNA extraction and genotyped together. This will reduce genotyping costs while still allowing the standard genomic prediction framework to operate on allele counts in the population as opposed to allele counts within a single individual. This strategy will allow for prediction of additive effects for genetic gain, as well as dominance effects to exploit population level heterosis.

To evaluate the efficacy of the proposed bulk genotyping method, we have obtained remnant seed from a diallel study that was previously evaluated for biomass in the field (Segovia-Lerma et al. 2004). These materials will be genotyped using a sequence based approach to estimate allele frequencies within each population, which will subsequently be used to estimate genetic relationships between populations. A genomic prediction model will be fit to evaluate the usefulness of the genotypic information to predict forage yield.

Segovia Lerma et al. (2004) created a half diallel by crossing all possible pairs of the nine historical varieties, and evaluated the resulting 36 hybrid populations along with the nine parental populations in the field in 1997 and 1998 near Las Cruces, NM in a replicated complete block design. Forage dry matter content data from this experiment for five harvests in each year was provided to us by Ian Ray at New Mexico State University (NMSU), as well as AFLP genotyping data from the nine parental populations (1544 AFLP markers, Segovia-Lerma et al. 2004).

Using the dominant AFLP markers (presence/absence), we constructed the hybrid population genotypes ‘in silico’ by summing the AFLP scores for each parental pair. Dominance scores were defined as 1 when one parent population had an allele that was missing in the other parent population, and 0 otherwise. This provided an approximation of the hybrid population genotypes that were subsequently used to calculate relationships between families and determine if the phenotypes were predictable using genotypic information alone. A ‘leave one family out’ strategy was implemented to validate that there is the genomic predictability in the somewhat small data set. The high level of genetic variability in the diallel was demonstrated by high genomic predictability, with leave one family out genomic predic-

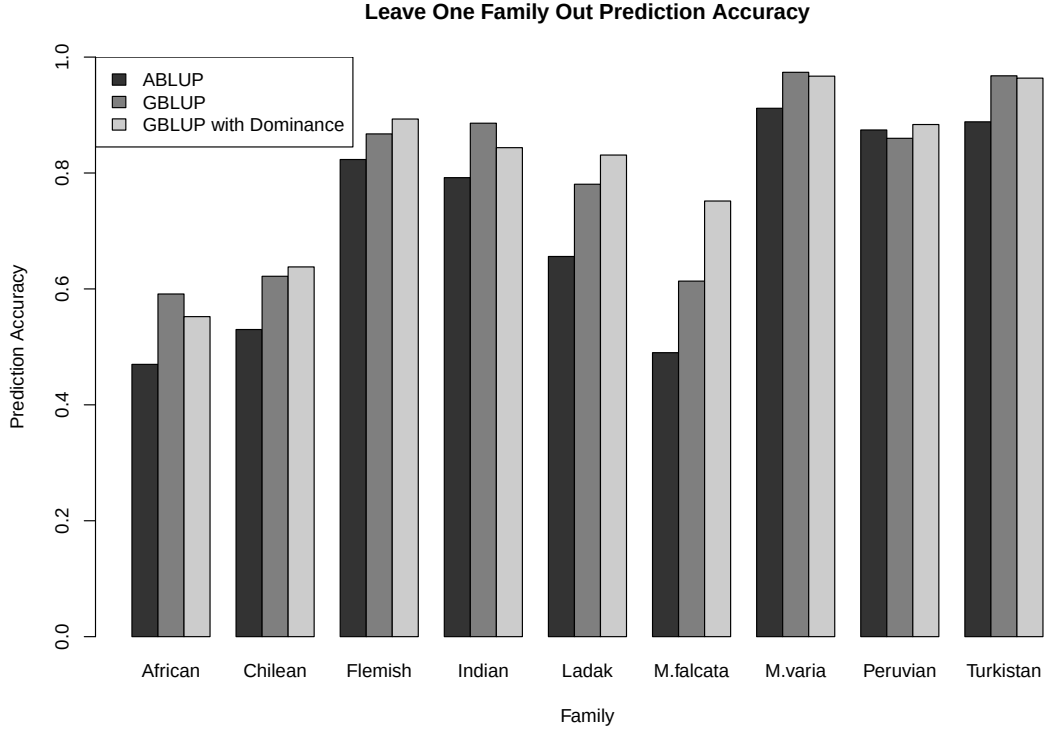


Figure 1: Prediction accuracy of each of the nine families using a leave one family out validation strategy. ABLUP, GBLUP and GBLUP with Dominance indicate models that use genetic relationships calculated from pedigree, additive marker scores and additive plus dominance marker scores, respectively.

tion accuracies ranging from 0.55-0.97 (Figure 1). Generally, inclusion of marker information increased prediction accuracy compared to a pedigree based prediction model. Inclusion of dominance predictors also increased accuracy for several families, demonstrating that dominance effects are important contributors, and will allow for prediction of hybrid vigor. This data set is atypical in the genetic signal due to the large genetic variability and known recent familial relationships, but we chose this population as a proof of concept because it is small (low cost), and has a high potential to detect differences (high genetic variance).

These results demonstrate that if a high-throughput genotyping platform can be built, then the population level genotyping strategy should be feasible for genomic prediction and selection in alfalfa, as well as other outcrossing synthetic populations. The genotyping technology used in that study was low throughput, and does not allow for estimation of allele dosage. We expect that allele dosage information should increase the predictive ability of such models, especially when the genetic variability is lower than in this diallel, as would be expected in elite breeding materials. Sequence information from these populations will allow us to target regions of the alfalfa genome to aid in building a high-throughput genotyping platform that can capture the allelic diversity of improved materials and sources of new genetic variability.

Research Plan

This is the first time our lab will use sequencing services from the BRC. We are requesting funds to bulk sequence remnant seed from the nine historical alfalfa populations from Segovia Lerma et al. (2004). Genotyping these nine populations using their sequences will allow us to answer initial questions about the proposed genotyping strategy for outcrossing synthetic populations.

To minimize cost and simulate the high-throughput that would be necessary for scaling up, DNA from each parent population will be bulk extracted by germinating a few hundred seeds. Then, 100 whole seedlings at the same developmental stage will be selected for bulk homogenization and DNA extraction. Here, we will use exactly the same number of seedlings for each population, but in practice, this will likely be ignored, as the expected allele frequencies are not influenced by the number of individuals, given the sample is sufficiently large. Two biological replicates will be conducted at the DNA extraction and library preparation stages to determine the variability in allele frequency estimates and guide method development. We could individually extract DNA and barcode each seedling, but this would drastically increase the labor and reagents by two orders of magnitude, with little benefit as phenotypes for individuals would not be available or practical.

DNA extraction and whole genome sequencing libraries would be prepared by BRC services. These nine populations will be combined with four hybrid populations from the diallel and eight Cornell varieties that we have obtained outside funding for sequencing. All 21 populations will be pooled and sequenced on five NextSeq500 lanes to obtain 30x coverage. Sequences will be aligned to the diploid alfalfa genome, CADL v0.95 ([Medicago HapMap project, NSF Project IOS-1237993](#), 2016). Allelic regions will be defined using coverage and polymorphism content, likely at protein encoding genes and other conserved regions. Alleles will be counted for each population using open source software constructed in house. This level of coverage should be sufficient to determine the minimal coverage and number of individuals per population necessary to obtain good allele frequency estimates.

The hybrid populations will be genotyped ‘in silico’ using the allele frequencies of the genotyped parental populations as

$$p_{a,k,ij} = \frac{1}{2}(p_{a,k,i} + p_{a,k,j}) \quad \forall i \neq j,$$

where $p_{a,k,i}$ and $p_{a,k,ij}$ is the i^{th} parent population allele frequency and the ij^{th} expected hybrid population allele frequency, respectively, for the a^{th} allele of the k^{th} marker. To validate the ‘in silico’ approach, seed from four of the hybrid populations will be bulk sequenced to compare observed and expected allele frequencies.

Dominant marker scores will also be constructed ‘in silico’ for the hybrid populations by using the product of the parental allele frequencies for all allele pairs at a single locus as

$$d_{aa',k,ij} = p_{a,k,i} \times p_{a',k,j} \quad \forall i \neq j \text{ and } \forall a \neq a'.$$

Allele frequencies and dominance scores will be used to calculate additive and dominance covariance relationships between hybrid and parental populations. Whole genome, random,

Table 1: Cost of reagents and sequencing.

Item	Number	Cost per unit	Total
DNA extraction (96 well plate)	1	\$550.00	\$550.00
TruSeq-equivalent WGS library preps (48)	1	\$2,851.00	\$2,851.00
MiSeq Nano library QC	1	\$300.00	\$300.00
NextSeq500, High Output 2 x 150 bp	1	\$4,791.00	\$4,791.00
Total			\$8,492.00

and selected subsets of loci will be sampled to evaluate predictive ability and necessary marker density. Genomic prediction accuracies of individual harvests, as well as annual yield of the the diverse diallel experiment will be compared to the adapted Cornell materials currently under evaluation in the field to determine what degree of genetic variability is necessary for adequate genomic prediction accuracy.

Looking forward, we will be collaborating with Breeding Insight to aid in construction of a high-throughput cost effective marker platform. The germplasm source materials sequenced here will allow for a wider survey of haplotypes which will aid imputation and construction of a Practical Haplotype Graph for alfalfa in BI. This data may also allow for determination of a “core” alfalfa genome that is highly conserved across elite and extant accessions of alfalfa, which may hold important genes for targeted amplicon marker design that can capture multiple alleles.

Budget

We are requesting that the BDF seed grant support DNA extraction, sequence library preparation and one lane of sequencing for these materials (total: \$8,492). We have recently obtained funding to sequence the four additional hybrid populations, as well as eight populations from the Cornell forage breeding program through another grant proposal detailed in the supplemental materials section. All data processing will be done in house to meet goals that differ from existing pipelines, as allelic regions will need to be defined to determine allele counts in each population. The software pipelines produced from this project will be made publicly available on GitHub. We expect the bioinformatics and analysis can be conducted in the course of one month from when the sequencing is completed.

Table 2: Time line of research.

Milestone	Expected Completion
DNA extraction & library preparation	November 2019
Sequencing	December 2019
Process sequences to produce allele counts	January 2019
Evaluate prediction ability	February 2019
USDA AFRI grant submission	March 2019
Manuscript preparation	April 2019

Supporting Documentation

This is a resubmission for the BDF Seed grant that was not funded in the last round of proposals in May 2019. We have made substantial progress toward building supporting evidence for a USDA-AFRI grant proposal (Agriculture and Food Research Initiative funding limit \$500,000; not to be confused with the Alfalfa Farmer Research Initiative mentioned below).

We were recently awarded a U.S. Alfalfa Farmer Research Initiative Checkoff Grant (https://www.alfalfa.org/pdf/USAFRI/USAFRI_RFP_2019_Round_1.pdf) in June 2019 to provide funds to genotype eight Cornell alfalfa varieties currently in the field in Geneva, NY. We have collected multi-spectral images every three to four days for two regrowth cycles in the summer of 2019, and will continue for three regrowth cycles in 2020. We have collected seed from all materials, and are currently working to process the imaging data. This award also included funds to sequence four hybrid populations from the diallel study described here, but we have yet to submit these materials for sequencing. Ideally, we plan to combine plant materials from this proposal with those from that grant award to be prepared and sequenced at the same time.

The two data sets will allow us to confirm the predictive ability of this genotyping strategy in both the highly diverse germplasm sources and adapted/elite materials. Comparison of the two sets of materials will also allow us to understand what loci are under fixation in NY materials compared to the genetic diversity of the germplasm pool.

Results from these two projects will be used as supporting information for a grant proposal to be submitted to USDA-AFRI in March 2020 (<https://nifa.usda.gov/funding-opportunity/agriculture-and-food-research-initiative-foundational-applied-science-program>), as previously mentioned on the cover page. That grant will be a collaborative effort between our lab, the Cornell forage breeding program under Don Viands and the NMSU alfalfa breeding program under Ian Ray. Briefly, materials in replicated trials recently establish in the field will be genotyped using the method proposed here, and all material will be imaged semiweekly to extract various vegetative indices and model yield as a function of genotype specific growth through time. That grant proposal will evaluate and begin to build a digital agriculture and genomic aided breeding initiative for forages in NY, NM and beyond.

References

- Barnes, DK et al. (1977). "Alfalfa germplasm in the United States: genetic vulnerability, use, improvement, and maintenance." In: *USDAARS Technical Bulletin* 1571, pp. 1–21.
- Segovia-Lerma, A et al. (2003). "AFLP-based assessment of genetic diversity among nine alfalfa germplasms using bulk DNA templates". In: *Genome* 46.1, pp. 51–58.
- Segovia-Lerma, A et al. (2004). "Population-based diallel analyses among nine historically recognized alfalfa germplasms". In: *Theoretical and applied genetics* 109.8, pp. 1568–1575.

Biographical Sketch: Kelly Robbins

Professional Preparation

BS	Animal Science <i>General Biology</i>	University of Tennessee	2002
MS	Animal Science <i>Breeding Genetics</i>	University of Georgia	2005
PhD	Animal Science <i>Statistical Genetics</i>	University of Georgia	2007

Appointments

- Cornell University: Assistant Professor, Plant Breeding and Genetics, 2017 – Present.
- Cornell University: Director, Genomic Open-source Breeding Informatics Initiative (GOBII) 2015 – 2017.
- Dow AgroSciences LLC: Quantitative Genetics Group Leader, 2011 – 2015.
- Purdue University: Adjunct, Dept. of Agronomy, 2010 – 2015.
- Dow AgroSciences LLC: Quantitative Geneticist, 2008 – 2011.

Five products most closely related to the proposal

1. Roorkiwal, M., Jarquin, D., Singh, M. K., Gaur, P. M., Bharadwaj, C., Rathore, A., Howard, R., Srinivasan, S., Jain, A., Garg, V., Kale, S., Chitikineni, A., Tripathi, S., Jones, E., **Robbins, K.**, Crossa, J., R. K. Varshney. 2018. Genomic-enabled prediction models using multi-environment trials to estimate the effect of genotype× environment interaction on prediction accuracy in chickpea. *Scientific Reports*, 8(1), 11701
2. Hickey, J. M., Chiurugwi, T., Mackay, I., Powell, W., Kilian, A., Jones, C., Canales, C., Grattapaglia, D., Bassi, F., Atlin, G., Gorjanc, G., Dawson, I., Rabbi, I., Ribaut, J., Rutkoski, J., Benzie, J., Lightner, J., Mwacharo, J., Parmentier, J., **Robbins, K.**, Skot, L., Wolfe, M., Rouard, M., Clark, M., Amer, P., Gardiner, P., Hendre, P., Mrode, R., Sivasankar, S., Rasmussen, S., Groh, S., Jackson, V., Thomas, W. and Y. Beyene. 2017. Genomic prediction unifies animal and plant breeding programs to form platforms for biological discovery. *Nature Genetics* 49:1297-1303.
3. Wang, Y., **Robbins, K. R.**, and R. Rekaya. 2010. Comparison of computational models for assessing conservation of gene expression across species. *PLoS ONE*. 5(10)
4. Wang, Y., Liu, X., **Robbins, K. R.**, and R. Rekeya. 2010. AntEpiSeeker: detecting epistatic interactions for case-control studies using a two-stage ant colony optimization algorithm. *BMC Research Notes*. 3:117
5. **Robbins, K. R.**, Zhang, W., R. Rekaya, and J. K. Bertrand. 2007. Ant colony optimization for feature selection in high dimensionality data sets. *Math. Med. Biol.* 24(4):413-26.

Five additional products

1. Boyd, N. L., **Robbins, K. R.**, Dhara, S. K., West, F. D., and S. L. Stice. 2009. Human Embryonic Stem Cell Derived Mesoderm-like Epithelium Transitions to Mesenchymal Progenitor Cells. *Tissue Engineering Part A*. 15(8): 1897-1907.
2. West, F. D., Machacek, D. W., Boyd, N. L., Pandiyan, K., **Robbins, K. R.**, and S. L. Stice. 2008. Enrichment and Differentiation of Human Germ-like Cells Mediated by Feeder Cells and Basic Fibroblast Growth Factor Signaling. *Stem Cell*. 26(11):2768-76.

3. Elias, A.A., **Robbins, K.R.**, Doerge, R.W. and Tuinstra, M.R. 2016. Half a Century of Studying Genotype× Environment Interactions in Plant Breeding Experiments. *Crop Sci* 56(5):2090-2105.
4. Rousselle, Y., Jones, E., Charcosset, A., Moreau, P., **Robbins, K.**, Stich, B., Knaak, C., Flament, P., Karaman, Z., Martinant, J., Fourneau, M., Taillardat, A., Romestant, M., Tabel, C., Bertran, J., Ranc, N., Lespinasse, D., Blanchard, P., Kahler, A., Chen, J., Kahler, J., Dobrin, S., Warner, T., Ferris, R., and S. Smith. 2015. Study on Essential Derivation in Maize: III. Selection and Evaluation of a Panel of Single Nucleotide Polymorphism Loci for Use in European and North American Germplasm. *Crop Sci.* 55.
5. Gupta, M., Dekelver, R. C., Palta, A., Clifford, C., Gopalan, S., Miller J., Novak, S., Desloover, D., Gachotte, D., Connell, J., Flook, J., Patterson, T., **Robbins, K. R.**, Rebar, E. J., Gregory, P. D., Urnov, F. D., and Petolino. 2012. Transcriptional activation of Brassica napus β -ketoacyl-ACP synthase II with an engineered zinc finger protein transcription factor. *Plant Biotechnology Journal.* 10(7).

Synergistic Activities

As lead of the Bioinformatics, Biometrics, and Data Management module of the Excellence in Breeding Platform (<http://excellenceinbreeding.org>), co-PI on a ARPA-E project “Integrated Seaweed Hatchery and Selective Breeding Technologies for Scalable Offshore Seaweed Farming”, and key personnel on the Genomic Open-source Breeding Informatics Initiative (GOBii, <http://www.gobiiproject.org>), the Robbins lab is actively engaged in research and implementation of genomic selection in multiple species for both breeding and pre-breeding applications.

Extensive project management experience acquired through 7 years working for a large multi-national seed company and as director of the GOBii project.

Biographical Sketch: Nicholas Santantonio

Professional Preparation

New Mexico State U.	Las Cruces, NM	Genetics	BS, 2010
New Mexico State U.	Las Cruces, NM	Plant and Env. Sciences	MS, 2013
Cornell U.	Ithaca, NY	Plant Breeding and Genetics	PhD, 2018
Cornell U.	Ithaca, NY	Quantitative Genetics	Postdoc, 2018 – present

Appointments

- Cornell University: T3 Data Curator, The Triticeae Toolbox, 2018

Five products most closely related to the proposal:

1. **Santantonio, N.**, Pierce, C.A., Steiner, R., Ray, I.M. 2018. Genetic Mapping of Water-Use Efficiency and Carbon and Nitrogen Metabolism in Drought-Stressed Alfalfa. *Crop Science*. 59(1) 92–106. <https://doi.org/10.2135/cropsci2018.05.0307>.
2. **Santantonio, N.**, Jannink, J.L. and M.E. Sorrells. 2019. Homeologous epistasis in wheat: the search for an immortal hybrid. *Genetics* (cover article). 211(3) 1105–1122. <https://doi.org/10.1534/genetics.118.301851>
3. Ray, I.M., Han, Y., Meenach, C.D., **Santantonio, N.**, Sledge, M.K., Pierce, C.A., Sterling, T.M., Kersey, R.K., Bhandari, H.S. and Monteros, M.J., 2015. Identification of Quantitative Trait Loci for Alfalfa Forage Biomass Productivity during Drought Stress. *Crop Science*, 55(5) 2012-2033. <https://doi.org/10.2135/cropsci2014.12.0840>
4. **Santantonio, N.**, Jannink, J.L. and M.E. Sorrells. 2019. Prediction of subgenome additive and interaction effects in allohexaploid wheat. *G3: Genes, Genomes, Genetics*. 9(3) 685–695. <https://doi.org/10.1534/g3.118.200613>.
5. **Santantonio, N.**, Jannink, J.L. and M.E. Sorrells. 2019. A low resolution epistasis mapping approach to identify chromosome arm interactions in allohexaploid wheat. *G3: Genes, Genomes, Genetics*. 9(3) 675–684 <https://doi.org/10.1534/g3.118.200646>.

Additional products:

1. **Santantonio, N.**. 2017. Bilinear. R package for fitting bilinear models (AMMI, GGE). GitHub. <https://github.com/nsantantonio/Bilinear>
2. Kissing Kucek, L., **Santantonio, N.**, Gauch, H., Dawson J., Mallory, E., Darby, H., and M.E. Sorrells. 2018. Genotype by environment interactions and local adaptations in organic wheat. *Crop Science*. 59(1) 25–32. <https://doi.org/10.2135/cropsci2018.02.0147>.
3. Veenstra L., **Santantonio, N.**, Jannink, J.L. and M.E. Sorrells. 2018. Influence of Genotype and Environment on Wheat Grain Fructan Content. *Crop Science*. 59(1) 190–198. <https://doi.org/10.2135/cropsci2018.06.0363>.

Synergistic Activities

US Alfalfa Farmer Research Initiative Grant (August 1, 2019 – July 31, 2020) *Evaluating Approaches to High-Throughput Phenotyping and Genotyping for Genomic Selection in Alfalfa*. Key personnel: project conception, grant proposal author, research lead. PI and Co-PIs: Kelly Robbins, Don Viands and Julie Hansen. National Alfalfa & Forage Alliance: \$36,811.

FAA Part 107 Remote Pilot Certification.

Thirteen years experience working in alfalfa, small grains, and onion breeding programs. Undergraduate and Master's research focused on the genetics and improvement of alfalfa productivity under drought for arid regions at New Mexico State University. PhD in small grains breeding at Cornell University combined bioinformatic and quantitative tools to determine the importance of duplicated genes to genomic prediction in wheat. Current work focuses on genetic simulations of breeding programs to optimize the use of genome-wide marker information for genetic gain and variety production. Strong background in applied plant breeding, quantitative genetics, statistics, programming and package development, bioinformatics, data management, polyploid genetics, and instruction.

CURRENT & PENDING SUPPORT

Name: Kelly Robbins

NAME (List/PD #1 first)	SUPPORTING AGENCY AND AGENCY ACTIVE AWARD/PENDING PROPOSAL NUMBER	TOTAL \$ AMOUNT	EFFECTIVE AND EXPIRATION DATES	% OF TIME COMMITTED	TITLE OF PROJECT
McCouch, Susan Jones, Elizabeth Buckler, Edward Mueller, Lukas Jannink, Jean-Luc Sorrells, Mark Sun, Qi Robbins, Kelly	Active: BILL & MEL FOUNDATION	\$18,498,113	11/21/2014- 10/30/2020	8.3%	DELIVERING HIGH-DENSITY GENOMICS BREEDER'S TOOLS
Lindell, Scott Yarish, Charles Kite- Powell, Hauke Smith, Bren Jannink, Jean-Luc Robbins, Kelly	Active: DOE ARPA-E	\$5,837,048	1/1/2018 – 12/31/2021	16.7%	INTEGRATED SEAWEED HATCHERY AND SELECTIVE BREEDING TECHNOLOGIES FOR SCALABLE OFFSHORE SEAWEED FARMING
Robbins, Kelly Mueller, Lukas	Active: CENTRO IN DE MEJORAMIENTO (GATES FDN)	\$3,519,246	11/1/2018- 10/30/2022	16.7%	EXCELLENCE IN BREEDING: BIOINFORMATICS AND DATA MANAGEMENT TOOLS AND SERVICES
Robbins, Kelly Santantonio, Nicholas	Active: NAFA	\$36,811	8/1/2019- 7/31/2020	4.2%	EVALUATING APPROACHES TO HIGH THROUGHPUT PHENOTYPING AND GENOTYPING FOR GENOMIC SELECTION IN ALFALFA
Buckler, Edward Coffman, Ronnie Gomez, Miguel Gore, Michael Robbins, Kelly Tufan, Hale	Active: USAID	\$29,000,000	9/25/2019- 9/24/2024	8.3%	FEED THE FUTURE INNOVATION LAB FOR CROP IMPROVEMENT

Kahn, Awais Robbins, Kelly	Pending:USDA-NIF/	\$500,000	1/1/2020- 12/31/2022	8.3%	DEVELOPING A GENOMIC RESOURCE & GENETIC FRAMEWORK TO IDENTIFY AND FINE-MAP FIRE BLIGHT RESISTANCE IN THE MALUS GERMPLASM REPOSITORY
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